

HARDWARE

TASK: 1

List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply.

Device	Type (SMPS / UPS)	Why it needs this power supply
Desktop Computer (CPU)	SMPS	Converts AC to stable DC power for internal components like motherboard, CPU, and GPU
LED TV	SMPS	Provides efficient and regulated power while reducing energy consumption and heat
Laptop Charger	SMPS	Converts high-voltage AC to low-voltage DC suitable for charging the laptop safely
Desktop Computer (with backup)	UPS	Prevents sudden shutdown during power cuts and protects data from loss
Wi-Fi Router (connected to UPS)	UPS	Keeps internet running during power outages for continuous connectivity

TASK: 2

Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections.

Hint: You can hand-draw and upload a photo, or use any free diagram tool like draw.io.



A: Input EMI Filtering A: Bridge Rectifier

B: input filter capacitors

Between B and C: primary side heat sink

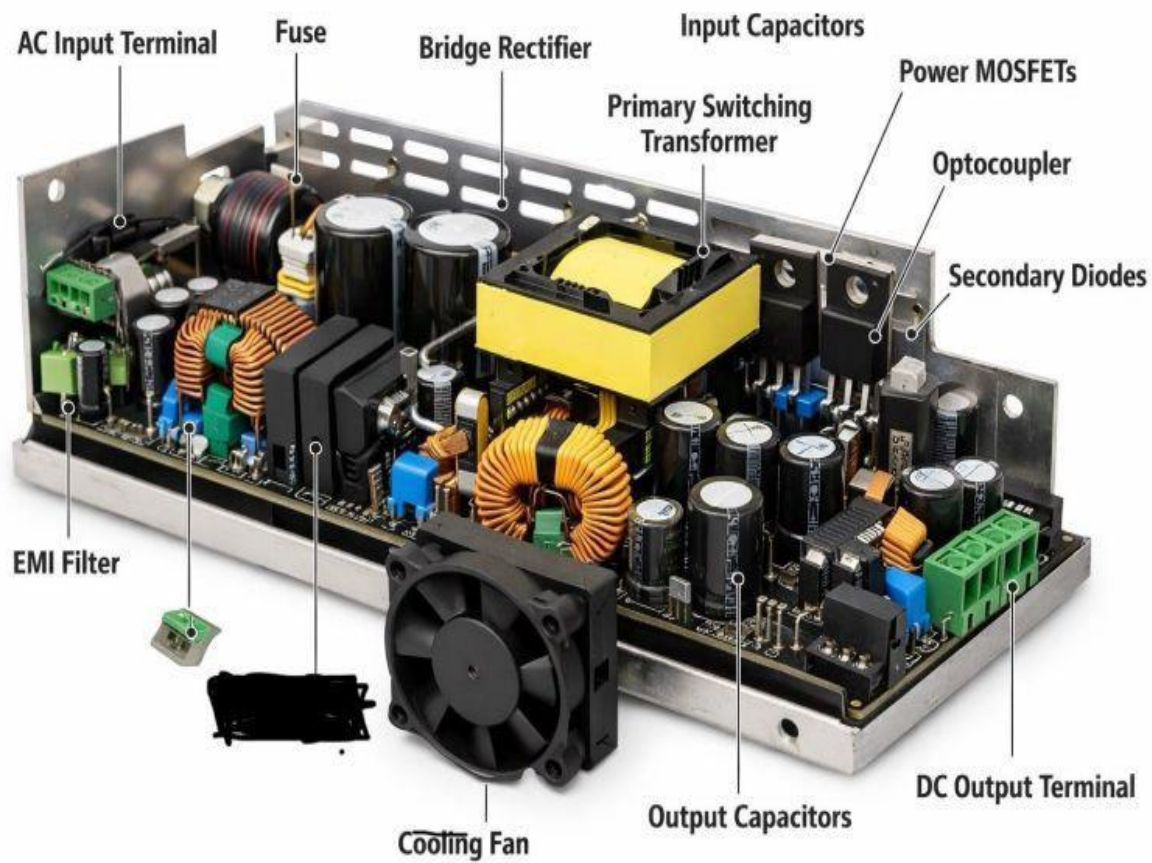
C: transformer

Between C and D: secondary side heat sink

D: output filter coil

E: output filter capacitors

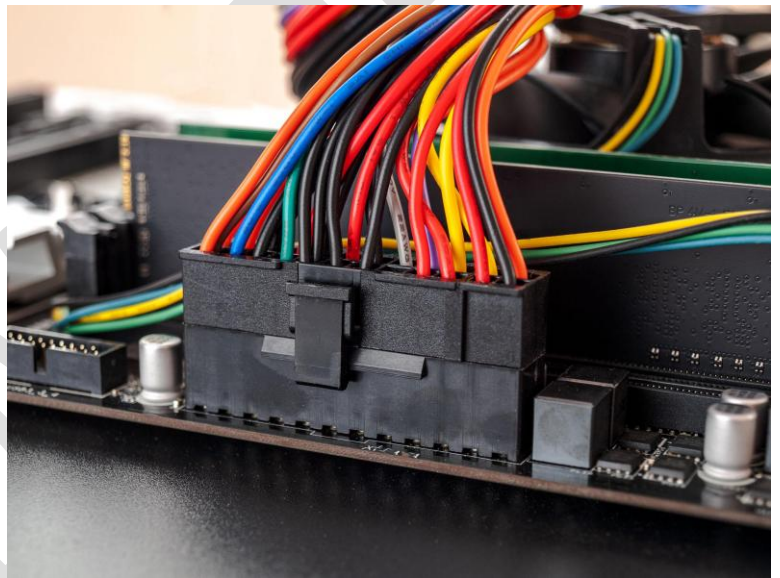
Interior view of an ATX SMPS



TASK: 3

Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.

1. ATX 24-Pin Motherboard Connector



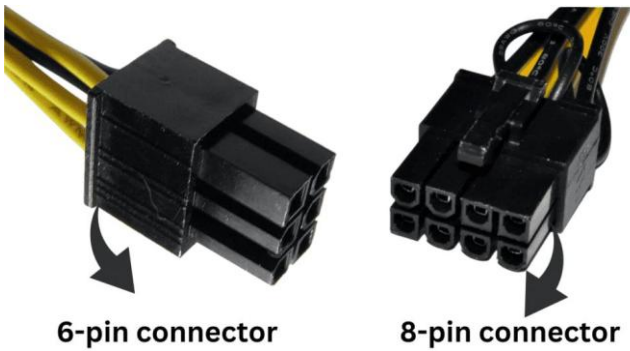
2. SATA Power Connector



3. Molex (4-Pin) Connector



4. PCIe 6/8-Pin Connector

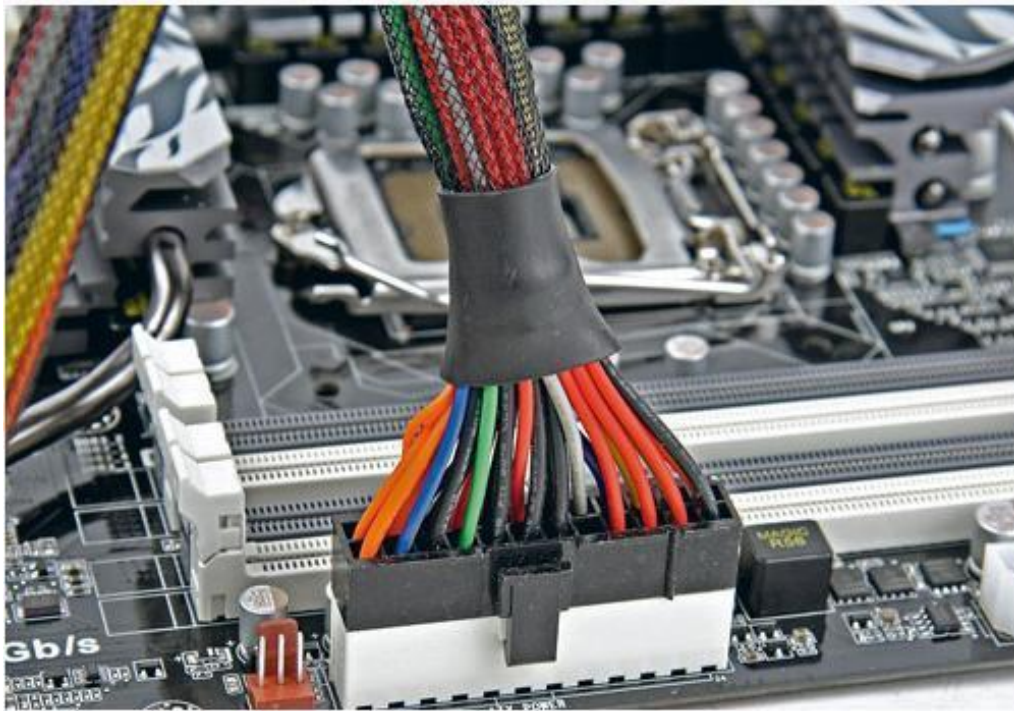


FOR BETTER KNOWLEDGE:

Connector	Powers	Key Safety Tip
ATX 24-pin	Motherboard	Match clip, don't force
SATA	HDD/SSD	Insert gently (L-shape)
Molex	Fans/old drives	Pull from connector
PCIe 6/8-pin	GPU	Fully connect all pins

TASK: 4

Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.

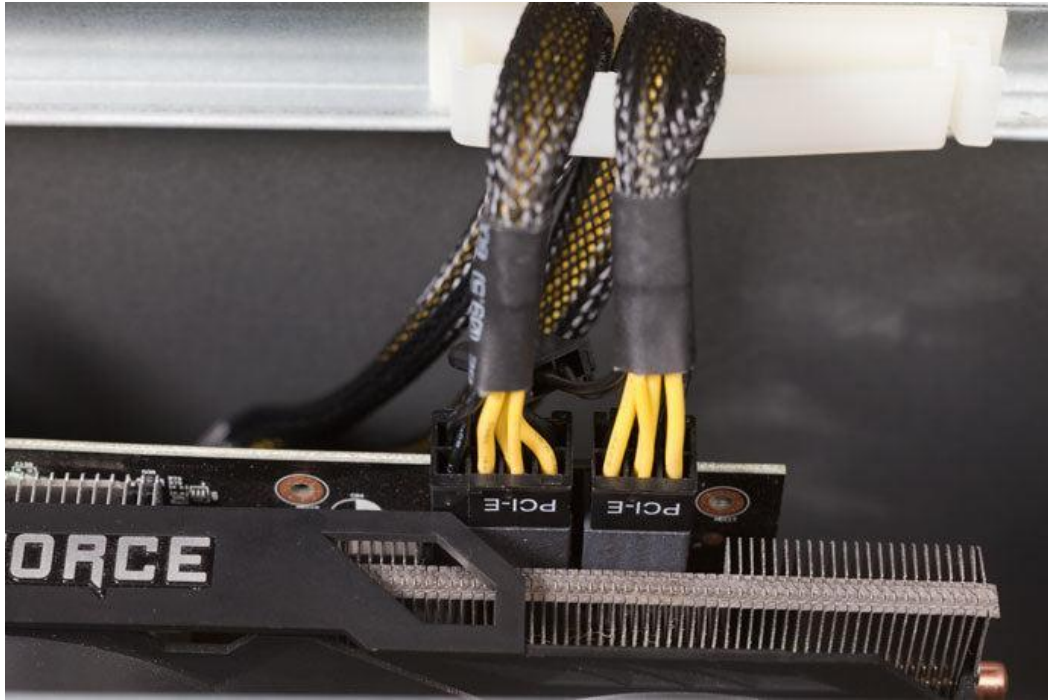


Steps:

1. Locate the **24-pin ATX cable** from the SMPS.
2. Find the **24-pin socket on the motherboard** (usually on the right side).
3. Align the clip and **press firmly until it clicks**.
4. Next, take the **8-pin CPU (EPS) connector** (sometimes 4+4 pin).
5. Plug it into the **CPU power socket** near the processor.

✓ Connectors Used:

- **24-pin ATX connector** → Main motherboard power
- **8-pin (4+4) EPS connector** → CPU power

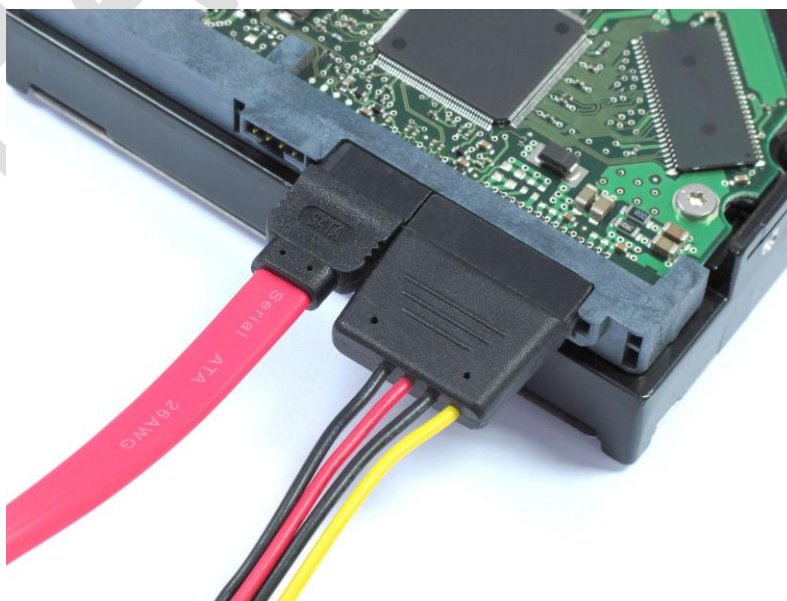


Steps:

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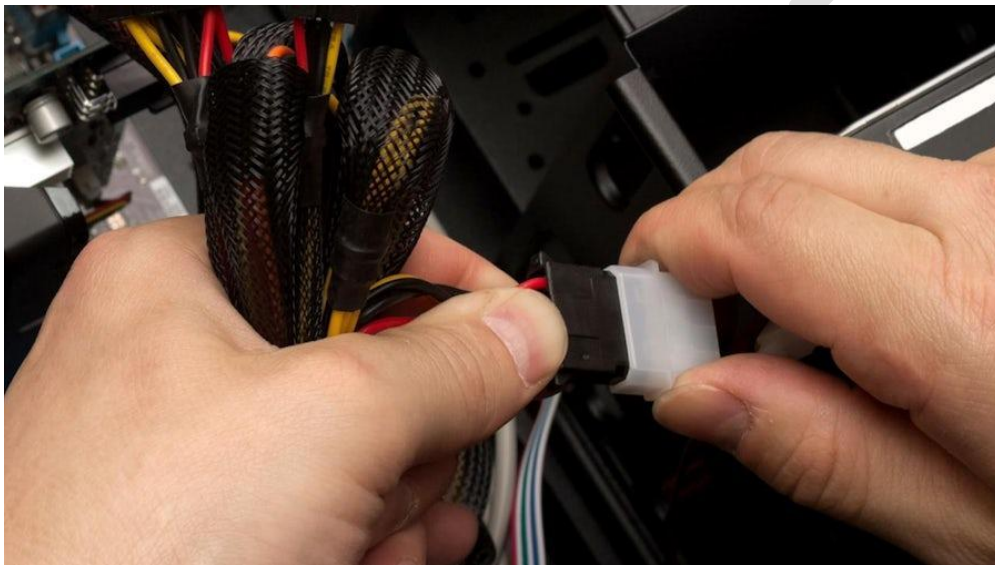


Steps:

1. Locate the **SATA power cable** from SMPS.
2. Connect it to the **SATA power port** on HDD/SSD.
3. Also connect a **SATA data cable** (from motherboard to drive).

✓ Connectors Used:

- **SATA power connector** → HDD/SSD/optical drives



Steps:

1. Use Molex cable if you have **older fans or accessories**.
2. Push the connector straight into the device port.

✓ Connectors Used:

- **Molex 4-pin connector** → Fans, older devices